

QUIZ

Write down the formula for mean squared error (mse) of estimator $\hat{\theta}_n$ that relates mse with $bias(\hat{\theta}_n)$ and $var(\hat{\theta}_n)$.

REVIEW OF (LAST PART OF) PREVIOUS LECTURE

Make statements like

with 95 % confidence $\mu \in (\bar{X}_n - m, \bar{X}_n + m)$

$$p = P(-k \leq Z \leq k) \Rightarrow k = \Phi^{-1}\left(\frac{1+p}{2}\right)$$

$$P\left(-k \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq k\right) = P\left(\bar{X}_n - \frac{k\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + \frac{k\sigma}{\sqrt{n}}\right)$$

Margin of error

CONFIDENCE INTERVALS

Ex: # of citations of a random paper is a random variable with $\sigma = 5$.

In a sample of 100 papers sample mean $\bar{X}_n$ is 7.5.

What is the 95 % confidence interval for mean μ ?

Step 1: $k = \Phi^{-1}\left(\frac{1+p}{2}\right) = \Phi^{-1}(0.975) = 1.96$

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Step 2: 95 % confidence interval for mean μ :

$$\left(\bar{X}_n - \frac{k\sigma}{\sqrt{n}}, \bar{X}_n + \frac{k\sigma}{\sqrt{n}}\right) = \left(7.5 - \frac{1.96 \times 5}{10}, 7.5 + \frac{1.96 \times 5}{10}\right) = (6.52, 8.48)$$

CONFIDENCE INTERVALS

For μ with unknown σ :

We have seen $\widehat{\sigma}_1^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is unbiased.

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Replace σ in **Step 2** with $\sqrt{\widehat{\sigma}_1^2}$.

Theorem: $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\widehat{\sigma}_1}$ has *t distribution* with $n - 1$ degrees of freedom.

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Theorem: $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\hat{\sigma}_1}$ has *t distribution* with $n - 1$ degrees of freedom.

Replace $k = \Phi^{-1}(\frac{1+p}{2})$ with $k = F_v^{-1}(\frac{1+p}{2})$ in Step 1.

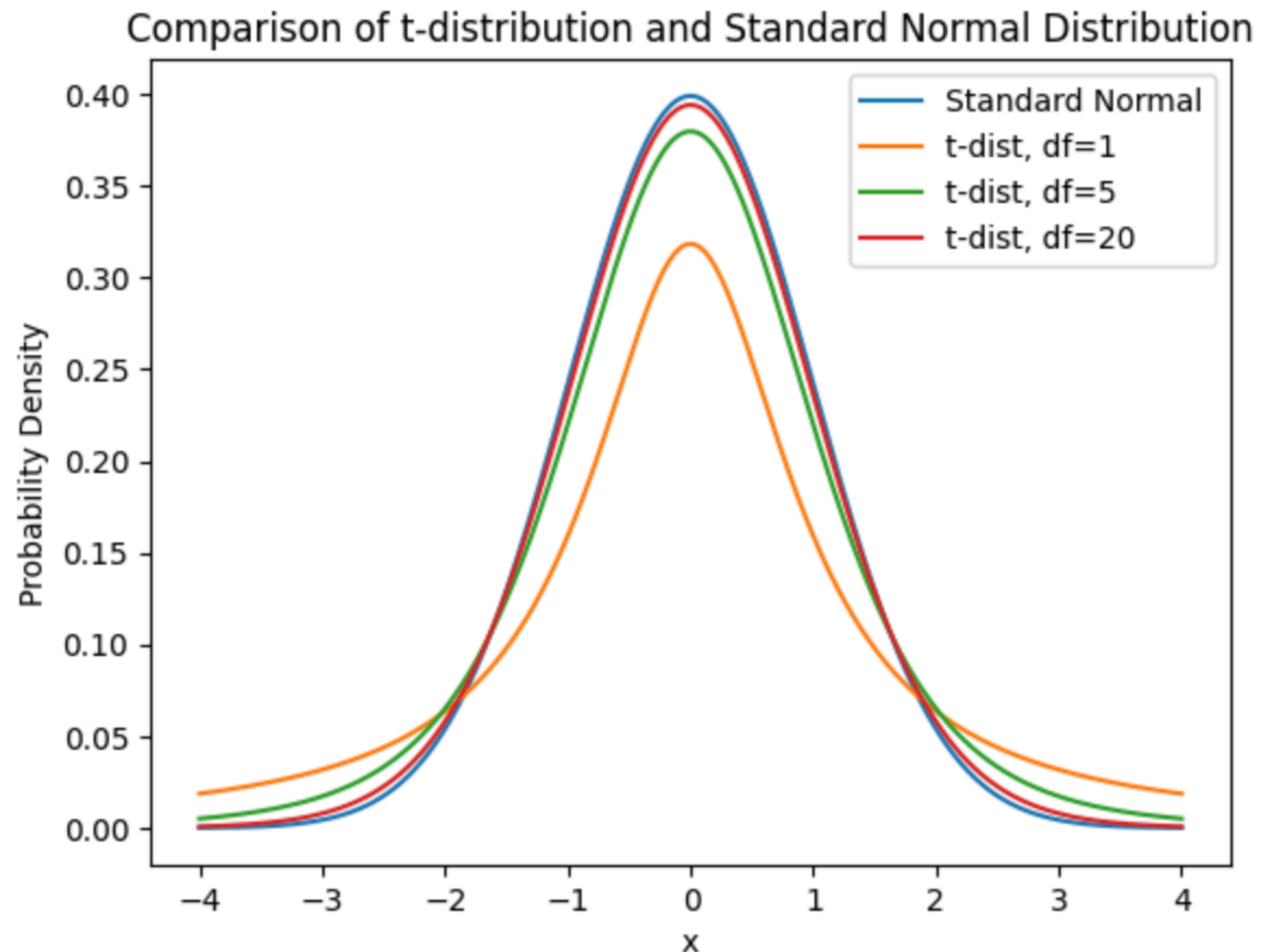
$v = n - 1$ is the degrees of freedom

CONFIDENCE INTERVALS

(Student's)
t distribution

How to find F_t^{-1} ?

$F_t^{-1}(1)$ with 20
degrees of freedom.



```
from scipy.stats import t  
t.cdf(1,20)
```

0.8353717114141455



William S Gosset
(aka *Student*)



While working at
Guinness

CONFIDENCE INTERVALS

Ex: Heights of men. Sample of size 9: 168, 176, 195, 182, 188, 150, 165, 170, 158

Find 95 % confidence interval for distribution mean.

CONFIDENCE INTERVALS

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Find $k = F_v^{-1}(\frac{1+p}{2})$ for $v = 8, p = 0.95$

$$k = F_8^{-1}(\frac{1+0.95}{2}) = F_8^{-1}(0.975) \approx 2.306$$

```
from scipy.stats import t  
t.ppf(0.975,8)
```

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$$\bar{X}_n = 172.4, \hat{\sigma}_1^2 \approx 206.03, \hat{\sigma}_1 \approx 14.35, \quad \text{Margin of error: } \frac{k\hat{\sigma}_1}{\sqrt{n}} \approx 11.03$$

$$\text{Confidence interval } (\bar{X}_n - \frac{k\hat{\sigma}_1}{\sqrt{n}}, \bar{X}_n + \frac{k\hat{\sigma}_1}{\sqrt{n}}) = (161.37, 183.43)$$

CONFIDENCE INTERVALS

How to interpret confidence intervals?

Not right to say:

μ is between 161.37 and 183.43 with 95% probability.

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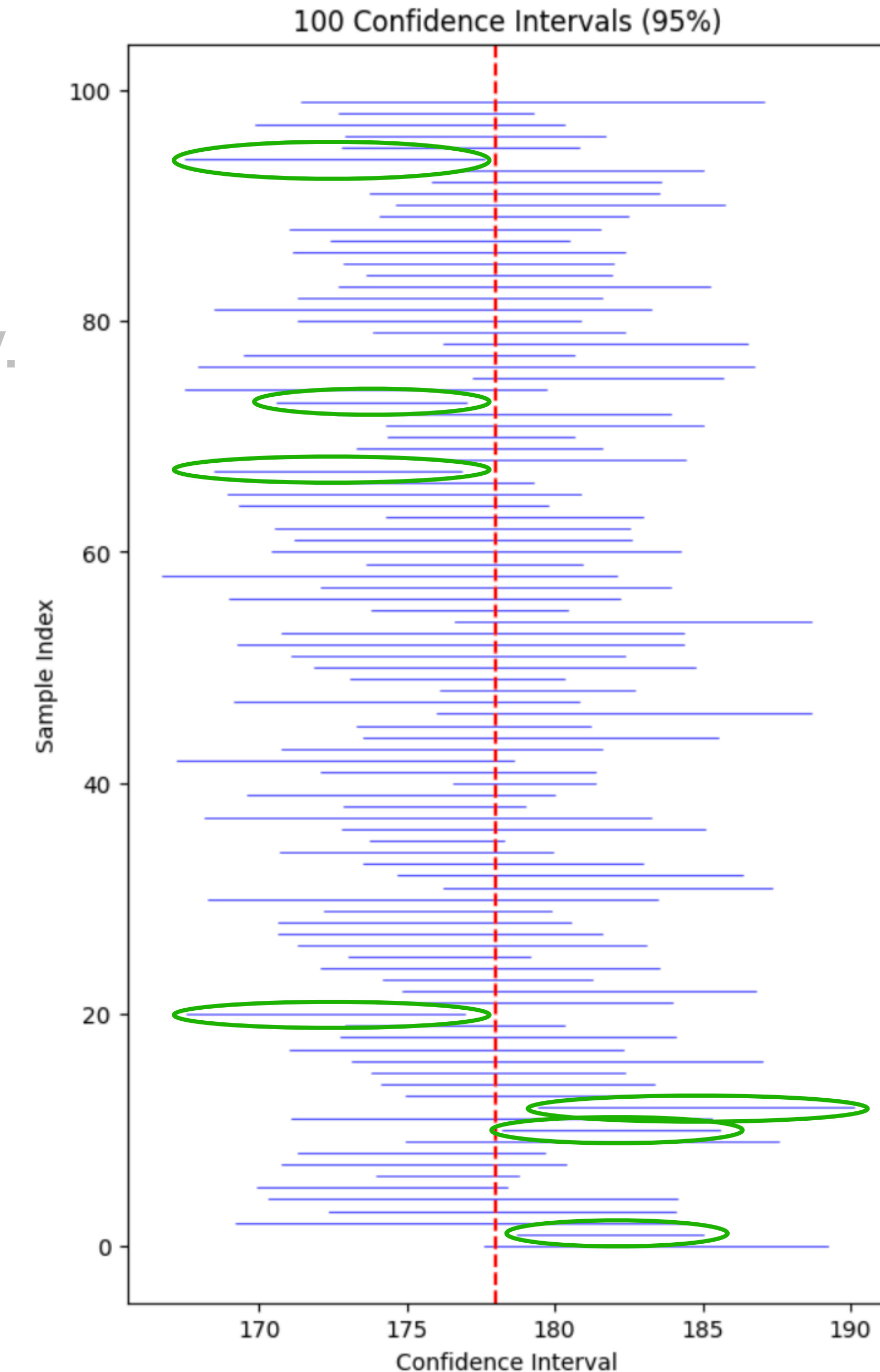
μ is between 161.37 and 183.43 with 95% probability.

100 observed 95 % confidence intervals based on samples of size 9, $\mu = 178$, $\sigma = 7$.

```
samples = [np.random.normal(178, 7, 9) for _ in range(100)]

# Calculate confidence intervals
intervals = []
for sample in samples:
    k = stats.t.ppf((1 + conf_level) / 2, df=sample_size - 1)
    s_mean = np.mean(sample)
    s_std = np.std(sample, ddof=1)
    mar_of_err = k * s_std / np.sqrt(sample_size)
    intervals.append((s_mean - mar_of_err, s_mean + mar_of_err))
```

For this sample 93 % of intervals contain $\mu = 178$.



Number of intervals outside 95% CI: 7

HYPOTHESIS TESTING

Statements about parameter/property θ of a distribution/population

Average GPA is < 2.8

Probability of heads of a coin is > 0.6

People eat more on weekends than in weekdays.

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$\theta = 3.2$ (simple), $\theta \in \{3, 4.2\}$ (composite), $\theta \in [0.5, 1.4]$ (composite)

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One-sided vs Two-sided

$\theta > 3.2$ (one-sided), $\theta < -1.5$ or $\theta > 2$ (two-sided), $\theta \neq 2$ (two-sided)

HYPOTHESIS TESTING

Two hypothesis

Null Hypothesis: H_0

Status quo, assumption believed to be true

Coin in my pocket, probability of heads $p = 0.5$

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Alternative Hypothesis: H_A

Complement of H_0

Novel finding after research

Coin has probability of heads $p \neq 0.5$

HYPOTHESIS TESTING

How to test?

Design experiment, collect data. Check:

If data shows strong evidence against H_0

Reject H_0 (In favor of H_A)

Else

Do not reject H_0 (Note: doesn't mean accept H_0)

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Analogy with the legal principle:

Presumed innocent (H_0) until proven guilty (H_A) with strong evidence against innocence.

HYPOTHESIS TESTING

More specifically,

Design experiment

Define **test statistic** T (related to hypothesis)

Distribution of T under H_0

Compute value t of the statistic T applied on the data

Reject H_0 if $t \in R$, for 'reasonable' **rejection region** R

HYPOTHESIS TESTING

Ex: Mean of a normal distribution with known variance

$X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with unknown μ . Test the hypotheses:

$$H_0 : \mu = \mu_0 \quad H_A : \mu \neq \mu_0$$

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Informally, reject H_0 if $\bar{X}_n$ far from μ_0 (strong evidence against H_0)

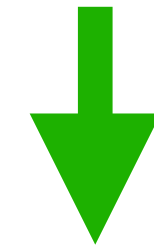
Formally, define test statistic, $T = |\bar{X}_n - \mu_0|$. Reject H_0 if $T \geq c$.

HYPOTHESIS TESTING

How to choose c ?

Significance level α :

$$P_{H_0}(T \geq c) \leq \alpha$$



Type I error:
We reject H_0
since $\bar{X}_n$ too
distant from μ_0
but H_0 is true.

Usually $\alpha = 0.05$

HYPOTHESIS TESTING

How to choose c ?

Significance level α : $P_{H_0}(T \geq c) \leq \alpha$

Reject H_0 if $\bar{X}_n$ lies
anywhere in shaded region.

Use quantile to find c .

